SYSTEMS ANALYSIS AND DESIGN / PHASE 1

Online Banking System

Functional and Non-functional Requirements

Elmira Orooji | Student ID 40173139

Instructor: Professor Jaderian

Ordibehesht 1404 (Solar Hijri)

In the name of God, the Most Compassionate, the Most Merciful.

Functional requirements

1. Authentication

1. Users must be able to authenticate at three levels, each with a distinct method.

1.1. At level 1, the system must authenticate users with their username and password.

1.2. At level 1, it must also grant access to corporate banking and the secure gateway.

1.3. At level 2, it must authenticate users with their card number, expiration date, CVV, and secondary card password.

1.4. At level 2, it must also grant access to the customer club.

1.5. At level 3, it must authenticate users with their national ID and mobile number.

1.6. At level 3, it must also grant access to Sayad cheque management.

1.7. It must provide a Back button at every step.

2. Password recovery

2. Users must be able to recover their password.

2.1. The system must provide a Forgot Password button.

2.2. It must display a mobile number field.

2.3. It must provide a Confirm button.

2.4. After confirmation, it must send a verification code.

2.5. After the code is entered, it must display fields for the new password and its confirmation.

2.6. It must update the user's password in the database.

2.7. It must provide a Back button at every step.

3. Fund transfers

3. Users must be able to transfer funds from their card to another person's card.

3.1. The system must provide a Fund Transfer option.

3.2. It must display a field for the recipient's card number or Sheba (Iranian IBAN).

3.3. Once it receives the information, it must offer a one-time password button for the online transfer.

3.4. It must request the one-time password from the originating bank.

3.5. It must verify the entered card number or Sheba against the database of Shetab member card numbers and valid bank Sheba numbers.

3.6. If validation succeeds, it must allow the user to proceed to final payment confirmation.

3.7. After final confirmation, it must report whether the transfer succeeded or failed.

3.8. It must provide a Back button at every step.

4. Transaction history

4. Users must be able to view completed financial transactions.

4.1. The system must provide a Financial Transactions menu option.

4.2. It must list completed transactions with their date, time, amount, and type: mobile top-up, mobile data, outgoing card transfer, incoming card transfer, or bill payment.

4.3. The menu must have Expenses and Income sections.

4.4. The Expenses section must provide a date-range selector.

4.5. It must show monthly expense totals as a bar chart.

4.6. It must calculate the amounts and percentages for fund transfers, top-ups, and bills and display them alongside each category.

4.7. The Income section must provide a date-range selector.

4.8. It must display the account balance for the last six months as a bar chart.

4.9. It must provide a Back button at every step.

5. Bill payments

5. Users must be able to pay different types of bills.

5.1. The system must provide a Store menu with icons for bills, mobile data, top-ups, loans, and Sayad cheques.

5.2. The Bills section must provide a New Bill button and let users choose water, electricity, gas, vehicle fine, mobile phone, or landline bills.

5.3. It must provide bill ID and payment ID fields, plus a phone number field for telephone bills.

5.4. It must provide a final confirmation button after the details are entered.

5.5. After final confirmation, it must display a success message or an error if payment fails.

5.6. It must provide a Back button at every step.

6. Mobile data purchases

6. Users must be able to buy a data bundle for any mobile operator.

6.1. The Store must offer a Mobile Data option with a mobile number field.

6.2. After the number is entered, it must list available bundles for that operator.

6.3. Selecting a bundle must open the bank payment page.

6.4. The system must accept the card number, CVV, expiration date, one-time password, and CAPTCHA.

6.5. It must validate the details.

6.6. It must report whether the purchase succeeded or failed.

6.7. It must avoid debiting the account if the purchase fails.

6.8. If the account was debited, it must refund the amount within 24 hours.

6.9. It must provide a Back button at every step.

7. Mobile top-ups

7. Users must be able to buy airtime for any mobile operator.

7.1. The Store must offer a Top-up option with a mobile number field.

7.2. After the number is entered, it must list available top-up amounts for that operator.

7.3. Selecting an amount must open the bank payment page.

7.4. The system must accept the card number, CVV, expiration date, one-time password, and CAPTCHA.

7.5. It must validate the details.

7.6. It must report whether the purchase succeeded or failed.

7.7. It must avoid debiting the account if the purchase fails.

7.8. If the account was debited, it must refund the amount within 24 hours.

7.9. It must provide a Back button at every step.

8. Loans

8. Users must be able to apply to the bank for a loan.

8.1. The Store must provide a Loans option.

8.2. It must show the bank's loan products, submitted applications, and their statuses.

8.3. It must show loans based on the user's financial behavior.

8.4. Each loan must have a guide explaining its amount, interest, guarantor requirements, and repayment terms.

8.5. The application form must include first and last name, comments, loan type, supporting documents, loan purpose, and digital signature.

8.6. The form must provide a button to upload supporting documents to the bank.

8.7. Upon submission, the system must validate each field.

8.8. For missing fields, it must display the appropriate error and ask the user to enter the missing information.

8.9. It must list the submitted application under submitted loans.

8.10. If the bank approves the loan, it must credit the loan amount to the user's balance.

8.11. It must display Successful for approved applications, Under Review for applications awaiting review, and Unsuccessful otherwise.

8.12. It must provide a Back button at every step.

9. Sayad cheque registration

9. Users must be able to register a Sayad cheque.

9.1. The Store must offer a Sayad Cheques option with All, Received Cheques, and Issued Cheques tabs.

9.2. The All tab must provide a New Operation button that displays a field for the Sayad cheque ID.

9.3. Entering the ID must enable the initially disabled Continue button.

9.4. The system must display fields for the cheque amount and due date.

9.5. It must enable Continue again.

9.6. It must display fields for the beneficiary's national ID and mobile number.

9.7. It must enable Continue again.

9.8. It must validate the beneficiary's national ID.

9.9. It must use the ID to look up and display the beneficiary's first and last name from the civil registry.

9.10. It must display fields for the cheque purpose and description.

9.11. The purpose list must include depositing collateral, customs duties, cash aid and charitable donations, foreign exchange, goods purchases, medical expenses, investment and stock exchange activities, and other government services.

9.12. The description field must ask the user to explain the selected purpose.

9.13. The system must display all previously entered information on a single page.

9.14. It must register the information after the user confirms it.

9.15. It must display cheque details in the Received Cheques and Issued Cheques tabs.

9.16. It must provide a Back button at every step.

10–11. User account

10. Users must be able to view their bank account details in the User Account section.

10.1. The Profile menu must offer a User Account option.

10.2. The system must display the account holder's name, opening date, account type, account number, and Sheba.

11. Users must be able to change their mobile number from the User Account section.

11.1. The Profile menu must offer a User Account option.

11.2. The Profile menu must offer a Change Mobile Number option.

11.3. The system must provide a field for the new number.

11.4. It must provide a button to confirm the entered number.

11.5. After confirmation, it must send a verification code to the user's number.

11.6. It must display a field for the code.

11.7. Once the verification code is entered, it must update the mobile number.

11.8. It must provide a Back button at every step.

12–13. Security and privacy

12. Users must be able to change their password.

12.1. The Profile menu must offer a Security and Privacy option.

12.2. The system must display fields for the current and new passwords.

12.3. It must replace the old password with the new one in the database.

13. Users must be able to sign out of their account.

13.1. The Profile menu must offer a Security and Privacy option.

13.2. The Sign Out option must end the user's session.

13.3. After sign-out, the system must show the username and password screen.

13.4. It must provide a Back button at every step.

14–15. Support

14. Users must be able to call support.

14.1. The Profile menu must offer a Support option.

14.2. Choosing Call must initiate a call to support.

15. Users must be able to access frequently asked questions.

15.1. Choosing Frequently Asked Questions must display the FAQ page.

15.2. The system must provide a Back button at every step.

16. Card blocking

16. Users must be able to block their card.

16.1. The system must provide a Card Services menu.

16.2. That menu must offer a Block Card option.

16.3. The system must request the card number, CVV, and expiration date.

16.4. It must provide a Submit button.

16.5. It must report whether the blocking operation succeeded or failed.

16.6. It must provide a Back button at every step.

17. Search financial reports

17. Users must be able to search financial reports by day or transaction title.

17.1. The Financial Reports menu must have a search section.

17.2. The system must display matching transactions when the user enters the first letter of the day or the transaction title.

17.3. It must display “No results found” when the search returns no matches.

Non-functional requirements

1. Usability (product requirement)

1. Users must be able to find every menu easily and without confusion.

1.1. The system must provide a simple, uncomplicated interface.

1.2. It must position and size buttons appropriately.

1.3. It must load pages in no more than two seconds.

1.4. It must make frequently used services easy to reach with minimal navigation.

1.5. It must provide services in as few steps as possible.

1.6. It must detect input errors and show an appropriate message. The Sayad cheque ID must have 16 digits. A password must contain at least eight characters, including uppercase and lowercase English letters and a digit.

1.7. It must size pages appropriately on all devices.

2. Accessibility standards (external requirement)

2. Users must be able to access online banking content in accordance with WCAG.

2.1. The system must present web content in accordance with WCAG so that people with disabilities, people with limited abilities, and older adults can access the same information and features as other users.

2.2. It must provide adequate spacing between lines and words.

2.3. It must allow users to zoom in and out.

3. Security (product requirement)

3. Users must be able to use online banking with confidence that their data is protected.

3.1. The system must use AES-256 encryption at rest and TLS 1.3 in transit.

3.2. It must use SHA-256 or SHA-3 hashing to verify data integrity.

3.3. It must use AWS Shield to mitigate attacks.

3.4. It must grant access only to authorized users whose identities have been verified.

4–6. Reliability (product requirements)

4. Users must be able to access the system 24 hours a day, seven days a week.

4.1. It must remain operational whenever users need to access account information or make transactions.

5. Users must be able to use the services without disruption from system failures.

5.1. On failure or other problems, the system must automatically add a backup system within one minute to maintain service access.

5.2. Recovery must be transparent to users and avoid service interruptions.

6. Users must be able to rely on correct processing of all financial transactions.

6.1. The system must implement transaction logs and rollback so that operations can be audited and reversed when necessary to preserve data integrity.

7. Scalability (product requirement)

7. Users must be able to stay connected and receive responses as the number of users and volume of data grow.

7.1. The system must handle 1,000,000 concurrent visits while maintaining optimal performance.

7.2. It must handle a 300% increase in traffic during holidays or promotional events without degradation in response time or availability.

7.3. It must use Microsoft SQL Server for its database to accommodate a large user base (organizational requirement).

8. Availability (product requirement)

8. Users must be able to access the system.

8.1. It must be available to users in Iran for 99.98% of business hours each month, measured in Iran local time.

8.2. It must operate continuously across devices and operating systems with 99% reliability.

8.3. It must check for potential problems under normal and abnormal operating conditions.

9. Compatibility (product requirement)

9. Users must be able to connect from common operating systems without disruption to functionality or user experience.

9.1. The system must run on Android, iOS, and Windows versions 7, 8, 9, 10, and 11, as listed in the source report.

9.2. It must behave and perform consistently across target operating systems.

9.3. It must support importing and exporting financial data in common formats such as CSV, XLSX, and QIF.

9.4. It must identify the internal systems with which it needs to interact.

9.5. It must be fully functional and maintain a consistent layout and behavior in major browsers, including Chrome, Firefox, and Safari.

10. Performance (product requirement)

10. Users must receive prompt responses when the system processes their information.

10.1. The system must return search results within three seconds.

10.2. It must ingest data streams from multiple sources at a rate of at least one million records per minute without data loss.